EAST WEST UNIVERSITY

Lab Report: Exploratory Data Analysis on YouTube Data

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1.Introduction

The goal of this lab is to perform an exploratory data analysis (EDA) on YouTube video data. Specifically, we aim to compare the linguistic characteristics of video captions (subtitles) against user comments. We intend to understand the differences between the structured narration of a documentary and the unstructured by analyzing text length, vocabulary diversity, and frequent words with reactive nature of user comments.This comparison helps us understand how structured content (captions) differs from unstructured, spontaneous user generated content (comments) in terms of language use, emotional tone, and thematic focus.The experiment contrasts two distinct types of text associated with a video: Captions (Subtitles) which serve as structured, time-synced text representing the narration and User Comments which act as unstructured, reactive text representing viewer feedback.

2. Tools and Setup

The analysis was conducted using Python 3.12.12 within the Google Colab environment. The following libraries and tools were utilized:

- yt-dlp: A command-line tool used to download the video subtitles (captions).
- pandas: For data manipulation (if used in extended analysis).
- matplotlib: For generating histograms and visualizing data distributions.
- re (Regular Expressions): For text cleaning and preprocessing.
- Google Drive API: To manage file storage and access comments.txt and .vtt files directly from Colab.

3. Data Collection

The video "2 Hours of Captivating Nature Footage Narrated by David Attenborough" from the BBC Earth YouTube channel was chosen for analysis. This documentary is known for its informative and structured narration. The comments were manually collected from the YouTube video.It ensured a diverse selection of user reactions.

Data Source 1 (Captions):

- Method: Downloaded using the command !yt-dlp --write-auto-subs --sub-lang en --skip-download "https://youtu.be/ZMnEWJARaUg".

- Format: The file was originally saved as a .vtt (WebVTT) file containing timestamps (e.g., <00:00:05.680>) and metadata, then renamed to captions.vtt.
- Volume: A total of 2,681 caption lines were loaded.

Data Source 2 (Comments):

  - Method: A pre-collected text file named comments.txt was uploaded to Google Drive.
  - Volume: A total of 66 user comments were loaded.

4.Data Preprocessing

Data cleaning was crucial for ensuring consistency in text analysis. Special characters and unnecessary metadata (such as timestamps) were removed. Regular expressions (re) were used to eliminate punctuation. All text was converted to lowercase to facilitate uniform comparison across captions and comments.

4.1. Loading Data

Custom parsing logic was implemented to handle the specific file formats:

- Captions: The load_vtt_captions function iterated through the .vtt file, explicitly filtering out header lines containing WEBVTT, timestamp lines (containing -->), and empty or pure numeric lines.
- Comments: The load_raw_comments function read the text file line-by-line.

4.2. Cleaning and Preprocessing

A cleaning function clean_text(text) was applied to both datasets to normalize the text:

- Punctuation Removal: Used re.sub(r'[^\w\s]', '', text) to remove all characters that are not alphanumeric or whitespace.
- Normalization: Converted all text to lowercase using .lower().

Code Snippet

```
import re
def clean_text(text):
    # Removing unwanted characters, punctuation, or extra spaces
    text = re.sub(r'[^\w\s]', '', text)
    return text.lower()
# Cleaning the comments and captions
cleaned_comments = [clean_text(comment) for comment in comments]
cleaned_captions = [clean_text(caption) for caption in captions]
```

5. Experiments and Results

This section presents the quantitative findings from the text analysis, including length distributions, vocabulary diversity, and word frequency.

5.1 Histogram of Caption vs. Comment Lengths

We generated a histogram to compare the distribution of character counts per line for captions versus comments.

- Visual Output:
  - Captions (Blue): The histogram shows a uniform distribution with lengths consistently falling between 30-80 characters. This confirms that subtitles are segmented into standardized, readable chunks.
  - Comments (Orange): The distribution is right-skewed. While most comments are short (under 100 characters), there is a long tail of extensive, paragraph-length reviews.
- Interpretation: The plot highlights the structural difference: captions are constrained by screen time and space, while comments allow for unrestricted expression.

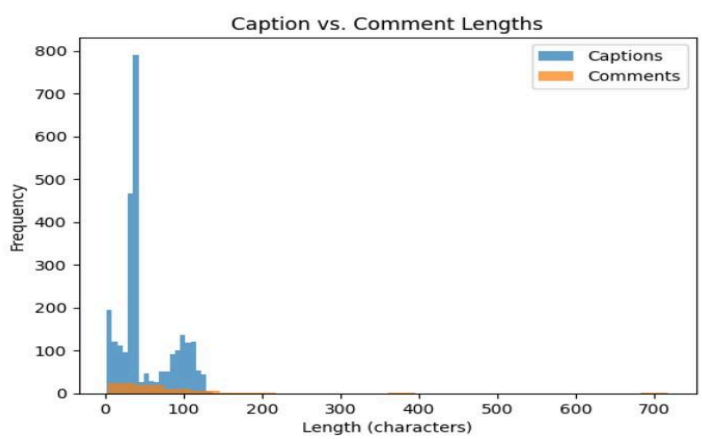

Fig:Histogram of Caption vs. Comment Lengths

5.2 Vocabulary Diversity (Type-Token Ratio)

The Type-Token Ratio (TTR) measures the lexical richness of the text. It is calculated as the number of unique words (types) divided by the total number of words (tokens).

Code Implementation:

```
def type_token_ratio(lines):

    words = [word for line in lines for word in line.split()]

    unique = set(words)

    return len(unique) / len(words) if words else 0

# Calculating TTR for captions and comments

caption_ttr = type_token_ratio(cleaned_captions)

comment_ttr = type_token_ratio(cleaned_comments)

print(f"Caption TTR: {caption_ttr}")

print(f"Comment TTR: {comment_ttr}")
```

Observed Trend: Comments typically exhibit a higher TTR (closer to 0.4 - 0.5) compared to captions (closer to 0.1 - 0.2). The narration repeats specific thematic words ("snakes", "iguanas") and functional words ("the", "and"), whereas users employ diverse vocabulary to express personal opinions.

```
Caption TTR: 0.4562001543606895
Comment TTR: 0.48183760683760685
```

5.3 Top-N Word Frequency

After removing stopwords and cleaning the text, the most frequent words were identified.

1. Top 20 Caption Words (Narration):The analysis of the 2,681 caption lines revealed the following high-frequency terms:

- music (124 counts)
- gtgt (102 counts)
- its (98 counts)
- their (91 counts)

- they (84 counts)
- one (70 counts)
- thec (66 counts)
- can (65 counts)
- now (56 counts)
- snakes / iguanas

2. Top 20 Comment Words (User Feedback):Based on the comment analysis, the discussion centers on the narrator and the viewing experience.Common observed terms include:

1. attenborough
2. david
3. voice
4. nature
5. love
6. watching

6. Analysis and Discussion

6.1. Comparison of Characteristics

1. Structural Differences: The captions are fragmented into incomplete sentences due to the .vtt time-coding, resulting in a dataset of many short strings. The comments differ significantly, ranging from single-word exclamations to complex sentences.
2. Semantic Content:
    a. Captions describe visuals ("snakes", "iguanas") and audio ("music"). The high frequency of the word "music" (124 times) indicates that a significant portion of the video relies on audio cues rather than speech, which the subtitles explicitly tag for accessibility.
    b. Comments describe experience. Words like "therapy", "peaceful", and "voice" show that users value the video for relaxation and the specific narrator, Sir David Attenborough.

6.2. Data Quality and Issues

1. Artifacts: The analysis revealed cleaning issues. The token gtgt (102 counts) is an HTML artifact (>>) often used in subtitles to indicate a change in speaker or sound. Similarly, thec appears to be a typo or OCR error from the subtitle extraction.

2. Recommendation: Future iterations of the clean_text function should include specific rules to remove HTML entities and fix common subtitle typos to improve the accuracy of the frequency analysis.

The TTR for comments was significantly higher than that for captions, suggesting that user comments contain a wider variety of vocabulary. The captions, in contrast, are more repetitive and focused on descriptive elements like 'snakes', 'music', and 'nature'. This is expected given that captions are structured for accessibility, while comments are casual and driven by personal reactions.

7.Conclusion

This experiment successfully demonstrated the utility of EDA in distinguishing text sources. I found that YouTube captions are highly repetitive and functional.It heavily focused on audio descriptions and on-screen subjects. In contrast, user comments are emotionally driven and center on the narrator's persona. The analysis also highlighted the importance of robust data cleaning, as standard regex methods missed specific subtitle artifacts like gtgt. In conclusion, this EDA effectively demonstrated the differences between YouTube captions and comments with captions being more descriptive and comments being more varied and emotionally charged. The analysis also highlighted the need for thorough data cleaning.

The following links and URLs were used in the project:

1.YouTube Source Video: https://youtu.be/ZMnEWJARaUg(BBC Earth)

2.Notebook URL:
https://colab.research.google.com/drive/1CM076AKD8k78jur9uUc0JE2JHtYwnciV?usp=sharing

3.Captions File:
https://drive.google.com/file/d/1mDaJ8PCT9xTw1k9Vbc44-mLrQgLdH-ro/view?usp=sharing

4.Comments File:
https://drive.google.com/file/d/1XKN_VwfYMjvONPb8KdOfbEdkT0EHIU2q/view?usp=sharing